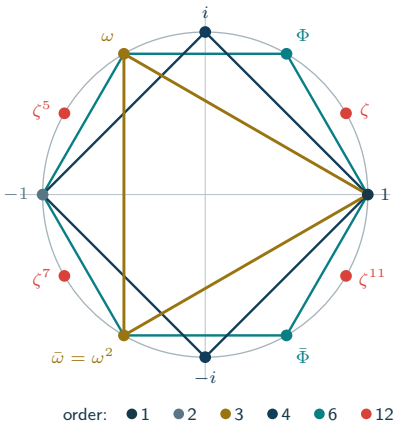
NUMBER THEORY ON ONE PAGE

Roots of unity of order 3, 4, 6 and 12

The symbols i, ω and Φ name roots of unity of orders 4, 3 and 6, and together they generate the 12th roots.

1 / THE RULE AND THE PICTURE

Orders 3, 4 and 6 sit inside order 12, and $12 = 4 \times 3$



$\zeta = \zeta_{12} = e^{i\pi/6}$; square μ_4, triangle μ_3, hexagon μ_6

The nth roots of unity are the corners of a regular n-gon on the unit circle, so the square ($n = 4$), the triangle ($n = 3$) and the hexagon ($n = 6$) all sit inside the 12-gon.
A corner is primitive when it generates its whole polygon, and the standard names are i for the square, $\omega = e^{2\pi i/3}$ for the triangle and $\Phi = e^{i\pi/3}$ for the hexagon.
The hexagon is the triangle with its sign, $\mu_6 = \mu_2 \times \mu_3$, so $\Phi = -\omega^2 = 1 + \omega$. Because 3 and 4 share no factor, every 12th root is a product of a fourth root and a cube root:

$$\mu_{12} = \mu_4 \times \mu_3, \qquad \zeta_{12} = -i\omega = i\bar{\Phi}$$

The square and the hexagon meet only in ± 1.

2 / THE SYMBOLS

Each root has an exact value, a polynomial and a field

root	angle	order	satisfies	field	ring
-1	$180°$	2	$x + 1$	$\mathbb{Q}$	$\mathbb{Z}$, two units
i	$90°$	4	$x^2 + 1$	$\mathbb{Q}(i)$	$\mathbb{Z}[i]$, four units
ω	$120°$	3	$x^2 + x + 1$	$\mathbb{Q}(\sqrt{-3})$	$\mathbb{Z}[\omega]$, six units
Φ	$60°$	6	$x^2 - x + 1$	$\mathbb{Q}(\sqrt{-3})$	$\mathbb{Z}[\Phi] = \mathbb{Z}[\omega]$, six units
ζ_{12}	$30°$	12	$x^4 - x^2 + 1$	$\mathbb{Q}(i, \sqrt{3})$	$\mathbb{Z}[\zeta_{12}]$, infinitely many units

The exact values are $i = e^{i\pi/2}$, $\omega = \frac{-1+i\sqrt{3}}{2}$, $\Phi = \frac{1+i\sqrt{3}}{2}$ and $\zeta_{12} = \frac{\sqrt{3}+i}{2}$.

3 / HOW THE SYMBOLS ARE DEGENERATE

Two kinds of ambiguity, one of sign and one of generator

Conjugate degeneracy. Each symbol names one of two roots of the same polynomial, namely $\pm i$, ω and $\bar{\omega} = \omega^2$, Φ and $\bar{\Phi}$, and algebra cannot tell the two apart. Once i is chosen, $\omega(i) = \frac{-1+i\sqrt{3}}{2}$ and $\Phi(i) = \frac{1+i\sqrt{3}}{2}$ are fixed (with $\sqrt{3} > 0$), and replacing i by $-i$ turns ω into $\bar{\omega}$ and Φ into $\bar{\Phi}$ together: one sign choice, not three.

Generator degeneracy. Since $\Phi = 1 + \omega$ and $\omega = \Phi - 1$, the symbols generate the same field and ring, $\mathbb{Q}(\omega) = \mathbb{Q}(\Phi)$ and $\mathbb{Z}[\omega] = \mathbb{Z}[\Phi]$, and adjoining i to either gives the same degree-four field,

$$\mathbb{Q}(\omega)(i) = \mathbb{Q}(\Phi)(i) = \mathbb{Q}(i, \sqrt{3}) = \mathbb{Q}(\zeta_{12})$$

So ω and Φ are interchangeable as generators of the field and different as numbers, of orders 3 and 6. The same holds for every odd n: $\mathbb{Q}(\zeta_{2n}) = \mathbb{Q}(\zeta_n)$.

4 / THREE QUADRATIC FIELDS INSIDE THE TWELFTH ROOTS

$\sqrt{3} = i(\bar{\Phi} - \Phi) = \zeta_{12} + \zeta_{12}^{-1}$ joins i and Φ

p	$\mathbb{Q}(i)$	$\mathbb{Q}(\sqrt{-3})$	$\mathbb{Q}(\sqrt{3})$	$\mathbb{Q}(\zeta_{12})$
5	split	inert	inert	2 primes of degree 2
7	inert	split	inert	2 primes of degree 2
11	inert	inert	split	2 primes of degree 2
13	split	split	split	4 primes, splits completely

The group $(\mathbb{Z}/12)^\times = \{1, 5, 7, 11\}$ has no element of order 4, so no prime stays inert in $\mathbb{Q}(\zeta_{12})$, and the quadratic characters multiply, $\chi_{-4}\chi_{-3} = \chi_{12}$, so every prime other than 2 and 3 splits in exactly one of the three fields or in all of them. This echoes $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{-2})$ and $\mathbb{Q}(i)$ inside $\mathbb{Q}(\zeta_8)$.

ASIDE / TRY THESE

Check $\zeta_{12} = -i\omega$: $(-i)\frac{-1+i\sqrt{3}}{2} = \frac{\sqrt{3}+i}{2}$. The primitive 12th roots are at $30°, 150°, 210°, 330°$.

LINKS TO OTHER SHEETS

The splitting, golden-type and cyclotomic sheets, and the $\sqrt{\pm 2}$ pair, which is the case $n = 8$.

Convention. μ_n the nth roots of unity; $\zeta = \zeta_{12} = e^{i\pi/6}$; $\bar{\omega} = \omega^2$; fates checked by factorising $x^2 + 1$, $x^2 + x + 1$ and $x^4 - x^2 + 1$ modulo p.

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